

Downtime, Dependency Failure, and Safe-Degradation Policy

Permitted partial operation when patient, retrieval, external, or model services fail

Document ID	Version	Effective	Review due
NB-BCP-001	1.0	2026-08-17	2027-08-17

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Owner: Business Continuity Lead
Approved by: Fictional Clinical Governance Committee
Classification: Training / synthetic corpus

Document control and retrieval metadata

Field	Value
Canonical title	Downtime, Dependency Failure, and Safe-Degradation Policy
Document ID	NB-BCP-001
Keywords	downtime, outage, dependency, OpenSearch, DynamoDB, openFDA, LLM, safe degradation
Authority	Business Continuity Lead
Applicability	Training simulations only
Relationship	Standalone extension; existing corpus documents remain unchanged

Retrieval rule

Retrieve whenever any application dependency times out, errors, returns malformed data, fails authentication, or produces an unverifiable response.

1. General rule

A failed dependency must never be represented as a negative clinical finding. 'No result returned' is not 'no risk found'. The system must disclose the failure and withhold conclusions that require that dependency.

2. Dependency matrix

Dependency failure	Permitted output	Prohibited output	Owner
Patient database / identity	Health/readiness status only	Any patient review or recommendation	Operations + data service
OpenSearch / policy RAG	Verified patient facts and existing signed order summary if allowed	New evidence-backed recommendation dependent on policy	Knowledge/platform team
openFDA	Patient/policy review with explicit missing drug-label evidence	Claim of no warnings/interactions	External integration owner
LLM provider	Deterministic facts, flags, status, and workflow next steps	Generated synthesis	AI platform owner
Citation validator	Uncited draft withheld from final display	FINAL_AUTHORIZED	Knowledge governance
Authorization service/data	Draft facts only	Final recommendation	Clinical governance
Langfuse/telemetry	Service may continue if audit/security controls remain	Dropping mandatory audit events	Observability owner
Authentication/access control	No protected processing	Patient data access	Security/identity owner

3. Failure classification

Class	Examples	Response
Transient	Timeout, rate limit, temporary network fault	Bounded retry with backoff/idempotency; then degrade safely
Data integrity	Malformed schema, inconsistent version, partial record	Reject response; quarantine/audit; human review
Authorization/security	Expired token, access denied, identity mismatch	Stop; no bypass; security workflow
Stale cache	Age beyond source-specific limit	Treat unavailable; do not silently use
Unknown	Unexpected exception or unclassified tool output	Fail closed for affected conclusion

4. Retry, caching, and recovery

- Retries are bounded, observable, and safe for the operation; no retry storm.
- Cache entries retain source, version, timestamp, patient binding, and expiry; sensitive caches are access-controlled.
- Expired or withdrawn policy content cannot be used even during downtime.
- After recovery, run readiness and integrity checks before resuming full service.
- Do not automatically replay medication-changing or external-communication actions.
- Reconcile partial audit events and identify affected requests.

5. User-facing message

Status: SYSTEM_UNAVAILABLE or [safe partial status]. Unavailable component: [plain description]. Completed safely: [verified portions]. Withheld: [affected conclusions]. Safest next action: [owner/contact/urgency]. Retry guidance: [if approved]. No unavailable check is described as normal.

6. Incident and communication

- Create incident ID and preserve timestamps, errors, affected services, request IDs, and data versions without unnecessary patient content.
- Notify clinical/operations owners when outage affects active safety workflows.
- Use an approved downtime process for urgent care; the application does not invent one.
- After restoration, review delayed tests, results, reconciliations, and recommendations to closure.

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